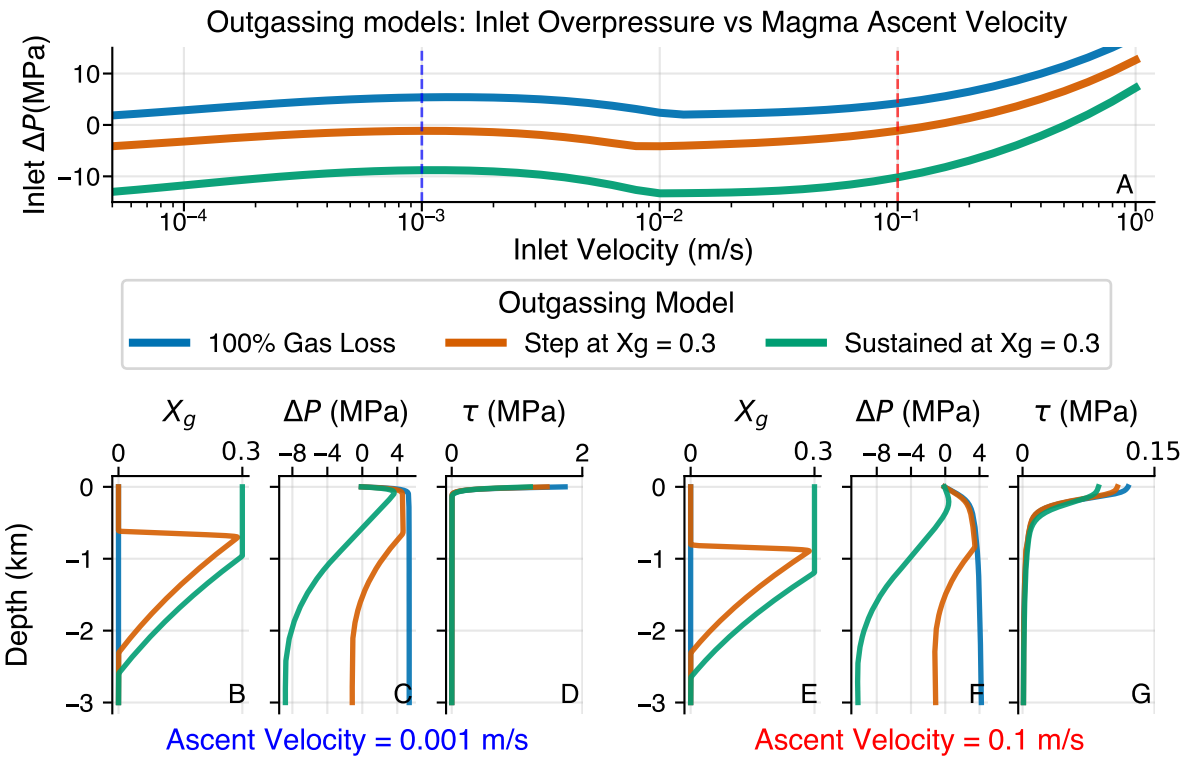
Outgassing models: Inlet Overpressure vs Magma Ascent Velocity
Inlet ΔP(MPa)
10
0
−10
A
10⁻⁴
10⁻³
10⁻²
10⁻¹
10⁰
Inlet Velocity (m/s)
Outgassing Model
100% Gas Loss
Step at Xg = 0.3
Sustained at Xg = 0.3
Xg
ΔP (MPa)
τ (MPa)
0
0.3
−8 −4 0 4
0
2
0.15
Depth (km)
0
−1
−2
−3
B
C
D
E
F
G
Ascent Velocity = 0.001 m/s
Ascent Velocity = 0.1 m/s